

ShuffleProofs.jl

Verificatum-compatible proofs of shuffle in Julia

Janis Erdmanis

- Verificatum proof → verify with **ShuffleProofs**
- ShuffleProofs proof → verify with **Verificatum**

Why Julia? HPC speed · rich type system · interactive REPL

The whole API on one slide

```
using CryptoGroups, ShuffleProofs

g = @ECGroup{P_192}()
verifier = ProtocolSpec(; g)
pk = g^42

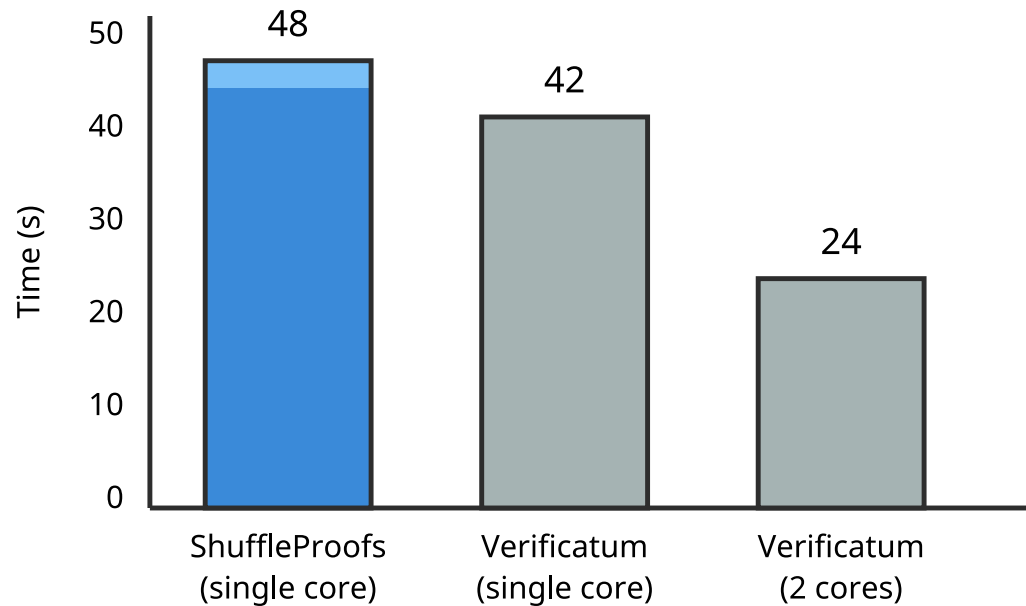
ciphertexts = [ElGamalRow(one(g), m) for m in [g, g^2, g^4]]

simulator = shuffle(ciphertexts, g, pk, verifier)

save(simulator.proof, proof_dir)
proof = load(VShuffleProof{typeof(g)}, proof_dir)

verify(simulator.statement, proof, verifier)
```

PoS Verification on P-256 (N = 100000)



Thanks!

ShuffleProofs.jl ·

github.com/PeaceFounder/ShuffleProofs.jl

janiserdmanis.org · peacefounder.org